

BML lecture #1: Bayesics

<http://github.com/rbardenet/bml-course>

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This year, the course takes place at Mines Paris; the room varies from week to week. Besides me, the lecturers will be

- ▶ Julyan Arbel (Inria Grenoble)
- ▶ Gabriel Victorino Cardoso (Mines Paris)



- 1** A warmup: Estimation in regression models
- 2** ML as data-driven decision-making
- 3** Subjective expected utility
- 4** Specifying joint models
- 5** 50 shades of Bayes

- ▶ *[...] practical methods for making inferences from data, using probability models for quantities we observe and for quantities about which we wish to learn.*
- ▶ *The essential characteristic of Bayesian methods is their explicit use of probability for quantifying uncertainty in inferences based on statistical data analysis.*
- ▶ *Three steps:*
 - 1 *Setting up a full probability model,*
 - 2 *Conditioning on observed data, calculating and interpreting the appropriate “posterior distribution”,*
 - 3 *Evaluating the fit of the model and the implications of the resulting posterior distribution. In response, one can alter or expand the model and repeat the three steps.*

- ▶ $y_{1:n} = (y_1, \dots, y_n) \in \mathcal{Y}^n$ denote observable data/labels.
- ▶ $x_{1:n} \in \mathcal{X}^n$ denote covariates/features/hidden states.
- ▶ $z_{1:n} \in \mathcal{Z}^n$ denote hidden variables.
- ▶ $\theta \in \Theta$ denote parameters.
- ▶ X denotes an $\mathcal{X}$ -valued random variable. Lowercase x denotes either a point in $\mathcal{X}$ or an $\mathcal{X}$ -valued random variable.

- ▶ Whenever it can easily be made formal, we write densities for our random variables and let the context indicate what is meant. So if $X \sim \mathcal{N}(0, \sigma^2)$, we write

$$\mathbb{E}h(X) = \int h(x) \frac{e^{-x^2/2\sigma^2}}{\sigma\sqrt{2\pi}} dx = \int h(x)p(x)dx.$$

Similarly, for $X \sim \mathcal{P}(\lambda)$, we write

$$\mathbb{E}h(X) = \sum_{k=0}^{\infty} h(k) e^{-\lambda} \frac{\lambda^k}{k!} = \int h(x)p(x)dx$$

- ▶ All pdfs are denoted by p , so that, e. g.

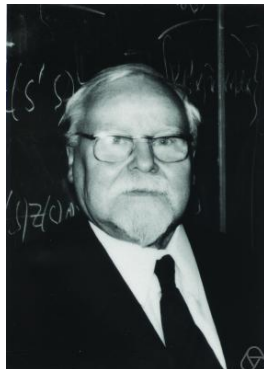
$$\begin{aligned}\mathbb{E}h(Y, \theta) &= \int h(y, \theta)p(y, \theta) dyd\theta \\ &= \int h(y, \theta)p(y, x, \theta) dx dy d\theta \\ &= \int h(y, \theta)p(y, \theta|x)p(x) dx dy d\theta\end{aligned}$$

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¹Germain et al., 2016.

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- ▶ A state space $\mathcal{S}$,
Every quantity you need to consider to make your decision.
- ▶ Actions $\mathcal{A} \subset \mathcal{F}(\mathcal{S}, \mathcal{Z})$,
Making a decision means picking one of the available actions.
- ▶ A reward space $\mathcal{Z}$,
Encodes how you feel about having picked a particular action.
- ▶ A loss function $L : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}_+$.
How much you would suffer from picking action a in state s .

- ▶ $\mathcal{S} = \mathcal{X}^n \times \mathcal{Y}^n \times \mathcal{X} \times \mathcal{Y}$, i.e. $s = (x_{1:n}, y_{1:n}, x, y)$.
- ▶ $\mathcal{Z} = \{0, 1\}$.
- ▶ $\mathcal{A} = \{a_g : s \mapsto 1_{y \neq g(x; x_{1:n}, y_{1:n})}, \quad g \in \mathcal{G}\}$.
- ▶ $L(a_g, s) = 1_{y \neq g(x; x_{1:n}, y_{1:n})}$.

PAC bounds; see e.g. (Shalev-Shwartz and Ben-David, 2014)

Let $(x_{1:n}, y_{1:n}) \sim \mathbb{P}^{\otimes n}$, and independently $(x, y) \sim \mathbb{P}$, we want an algorithm $g(\cdot; x_{1:n}, y_{1:n}) \in \mathcal{G}$ such that if $n \geq n(\delta, \varepsilon)$,

$$\mathbb{P}^{\otimes n} \left[\mathbb{E}_{(x,y) \sim \mathbb{P}} L(a_g, s) \leq \varepsilon \right] \geq 1 - \delta.$$

▶ $\mathcal{S} =$

▶ $\mathcal{Z} =$

▶ $\mathcal{A} =$

▶

► $\mathcal{S} =$

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The subjective expected utility principle

- 1 Choose $\mathcal{S}, \mathcal{Z}, \mathcal{A}$ and a loss function $L(a, s)$,
- 2 Choose a distribution p over $\mathcal{S}$,
- 3 Take the the corresponding Bayes action

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s \sim p} L(a, s). \quad (1)$$

Corollary: minimize the posterior expected loss

Now partition $s = (s_{\text{obs}}, s_{\text{u}})$, then

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s_{\text{obs}}} \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

In ML, $\mathcal{A} = \{a_g\}$, with $g = g(s_{\text{obs}})$, so that (1) is equivalent to $a^* = a_{g^*}$, with

$$g^*(s_{\text{obs}}) \triangleq \arg \min_g \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

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- ▶ PGMs (aka “Bayesian” networks) represent the dependencies in a joint distribution $p(s)$ by a directed graph $G = (E, V)$.
- ▶ Two important properties:

$$p(s) = \prod_{v \in V} p(s_v | s_{\text{pa}(v)}) \quad \text{and} \quad y_v \perp y_{\text{nd}(v)} | y_{\text{pa}(v)}.$$

Also good to know how to determine whether $A \perp B | C$; see (Murphy, 2012, Section 10.5).

***d*-blocking**

An undirected path P in G is *d*-blocked by $E \subset V$ if at least one of the following conditions hold.

- ▶ P contains a “chain” $a \rightarrow b \rightarrow c$ and $b \in E$.
- ▶ P contains a “tent” $a \leftarrow b \rightarrow c$ and $b \in E$.
- ▶ P contains a “v-structure” $a \rightarrow b \leftarrow c$ and neither b nor any of its descendants are in E .

Theorem

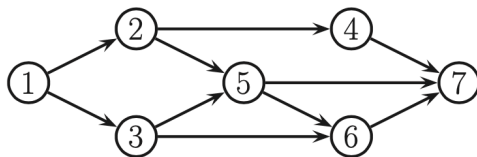


Figure 10.11 A DGM.

- ▶ Does $x_2 \perp x_6 | x_5, x_1$?
- ▶ Does $x_2 \perp x_6 | x_1$?
- ▶ Write the joint distribution as factorized over the graph.

Let us take a little test!

- ▶ Today's test will not count towards the final grade, I am benchmarking Wooflash.
- ▶ Please sign up with an academic email.



Theorem (Barron, 1997)

Assume $(X_i, Y_i) \sim P^$ i.i.d., and let P be described by drawing θ from a prior, then (X_i) from the first marginal of P^* , and, conditionally on (X_i) and θ , drawing each Y_i independently from $P_\theta(\cdot|X_i)$. Then under a mild condition on the prior, for all n ,*

$$\mathbb{E}_{P^*} \frac{1}{n} \sum_{i=1}^n [\text{KL}(P^* || P(\cdot|X_{1:i-1}, Y_{1:i-1})) - \text{KL}(P^* || P_{\tilde{\theta}})] = \mathcal{O}(\log n/n).$$

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An issue (or is it?)

Depending on how they interpret and how they implement SEU, you will meet many types of Bayesians (46656, according to Good).

A few divisive questions

- ▶ Using data or the likelihood to choose your prior; see Lecture #5.
- ▶ Using MAP estimators for their computational tractability, like in inverse problems

$$\hat{x}_\lambda \in \arg \min \|y - Ax\| + \lambda \Omega(x).$$

- ▶ When and how should you revise your model (likelihood or prior)?
- ▶ MCMC vs variational Bayes (more in Lectures #2 and #3)

- [1] A. Gelman et al. *Bayesian data analysis*. 3rd. CRC press, 2013.
- [2] P. Germain et al. "PAC-Bayesian theory meets Bayesian inference". In: *Advances in Neural Information Processing Systems* 29 (2016).
- [3] K. Murphy. *Machine learning: a probabilistic perspective*. MIT Press, 2012.
- [4] S. Shalev-Shwartz and S. Ben-David. *Understanding machine learning: From theory to algorithms*. Cambridge university press, 2014.